

Madeline Brooks, PhD, MPH

New Haven, CT | (302) 562-0427 | madeline.brooks42@gmail.com | <https://www.linkedin.com/in/madelinebrooks/>

EDUCATION

- 08/2022 – 08/2026 **PhD, Epidemiology**, Johns Hopkins University
- 09/2018 – 05/2019 **Certificate, Spatial Analysis for Public Health**, Johns Hopkins University
- 08/2016 – 08/2018 **Master of Public Health, Public Health Practice**, Thomas Jefferson University
- 08/2011 – 12/2014 **Bachelor of Arts, Mass Communication**, University of Delaware

SELECTED COURSEWORK

Advanced Theory and Methods in Epidemiology; Advanced Methods for the Design and Analysis of Cohort Studies; Analysis of Multilevel and Longitudinal Data; Causal Inference in Medicine and Public Health I; Study Design and Analysis for Causal Inference with Time-Varying Exposures; Statistics for Psychosocial Research: Measurement

PROFESSIONAL EXPERIENCE

- 09/2026 – Present **Postdoctoral Associate**, Public Health Data Science and Data Equity, Yale University
- Conduct methods development and applied research using electronic health records and real-world healthcare data
- 08/2023 – 08/2026 **Graduate Research Assistant**, NA-ACCORD, Johns Hopkins University
- Developed and evaluated traditional regression and machine learning models (including DeepSurv), using SHAP values to quantify and rank predictors of virologic failure among people with HIV
 - Conducted data management, statistical modeling, and visualization using Python and R
- 12/2022 – 08/2026 **Graduate Research Assistant**, ECHO, Johns Hopkins University
- Employed clustering algorithms and ensemble learning to identify neighborhood-level social phenotypes and validate them using spatial regression (SAR, SEM) models
 - Developed reproducible code to create datasets of social and environmental exposures for use in multiple studies
- 03/2021 – 06/2022 **Research Investigator**, Institute for Research on Equity and Community Health, ChristianaCare
- Conducted studies of population health and health care delivery using electronic health record data
 - Operated the Geospatial Analytics Core, using geographic information systems and spatial statistics to understand distributions of patient health outcomes and their place-based social determinants
 - Characterized the distribution of triple negative breast cancer and its risk factors within New Castle County to guide geographically targeted cancer control and prevention interventions
 - Partnered with the New Castle County government to create reports informing and evaluating the placement of COVID-19 testing sites in New Castle County
- 08/2018 – 02/2021 **Research Associate**, Institute for Research on Equity and Community Health, ChristianaCare
- Led a research project to describe the standard of nutrition-focused discharge care instructions provided to malnourished adult inpatients
 - Conducted quantitative and qualitative analyses to validate the accuracy of race, ethnicity, and language data collected by registrars
- 10/2016 – 04/2018 **Work-Study**, Office of Institutional Research, Thomas Jefferson University

- Designed alumni surveys and assembled data to monitor key performance measures for academic programs
- 01/2017 – 09/2017 **Research Student**, Jefferson College of Nursing, Thomas Jefferson University
- Analyzed 11 years of qualitative survey data from more than 180 nursing students who completed a clinical rotation in a maximum-security prison
 - Developed a manuscript and poster discussing the value of prison-based clinical rotations for nursing and public health practice
- 01/2017 – 02/2017 **Research Student**, Enhancing Services for Homeless Populations, Thomas Jefferson University
- Worked with an interdisciplinary team to pilot virtual role-play simulations designed to teach health professions students best practices for working with individuals experiencing homelessness
- 06/2016 – 02/2017 **Intern**, Delaware Academy of Medicine/Public Health Association
- Wrote a manuscript on social support interventions for diabetes management and created resource directories on violence prevention and health data collection
- 01/2015 – 03/2016 **Communications Specialist**, AstraZeneca HealthCare Foundation
- Drafted 3 manuscripts for submission to academic journals; developed a poster on public health grant-making for the American Public Health Association's 2016 Annual Meeting
 - Planned and executed 4 roundtable discussions on chronic disease programming with health care and nonprofit leadership

SOFTWARE SKILLS

R/RStudio, Stata, SAS, Python, GitHub, RShiny, REDCap, ArcGIS, SaTScan

AWARDS & ACTIVITIES

- 02/2026 – 08/2026 **NIH Ruth L. Kirschstein National Research Service Award (F31)**, National Institutes of Mental Health (NIMH), NIH
- 1F31MH142332-01 – “Identifying intervention opportunities in the continuum of cognitive impairment among persons aging with HIV”
- 07/2025 – 02/2026 **T32 Training Grant for Epidemiology and Biostatistics of Aging**, Johns Hopkins University
- Terminated T32 support early due to receipt of F31 award
- 07/2025 – 08/2026 **Training and Mentorship Chair**, Epidemiology Student Organization, Johns Hopkins University
- 07/2024 – 08/2026 **Co-Chair**, Social Epidemiology Student Organization, Johns Hopkins University
- 12/2025 **Poster Session Award**, Research Symposium on Engineering in Healthcare, Johns Hopkins University
- 06/2025 **Certificate of Completion**, Advanced Confounding Adjustment, CausaLab, Harvard T.H. Chan School of Public Health
- 05/2025 **Leon Gordis Centennial Scholarship**, Epidemiology Department, Johns Hopkins University
- In recognition of excellence as a department teaching assistant
- 08/2024 – Present **Curriculum Committee Representative**, Epidemiology Student Organization, Johns Hopkins University
- 01/2024 – 05/2024 **Interim Service Chair**, Epidemiology Student Organization, Johns Hopkins University
- 06/2023 – 05/2024 **Secretariat**, Epidemiology Student Organization, Johns Hopkins University

01/2020 – 06/2022	Instructor , English Conversation Program, Newark United Methodist Church
03/2019	Selection Committee Member , Delaware INBRE Summer Scholars Program
07/2017 – 06/2018	President , Jefferson Student Association for Public Health Education
06/2018	Student Service Award , Jefferson College of Population Health

SERVICE & TEACHING

08/2025 – 12/2025	Teaching Assistant , Doctoral Seminar in Epidemiology Guided second-year doctoral students in the development of dissertation proposals
08/2025 – 10/2025 08/2024 – 10/2024	Teaching Assistant , Advanced Methods for Design & Analysis of Cohort Studies Provided individualized teaching support and lab grading for an advanced epidemiologic methods course covering parametric survival analysis, mixed-effects models, and competing risks
03/2025 – 05/2025	Teaching Assistant , Epidemiologic Methods for Epi Doctoral Students IV Mentored first-year doctoral students as they developed independent research projects
01/2025 – 03/2025	Teaching Assistant , Epidemiologic Methods III Provided instruction in weekly labs and office hours covering regression, survival analysis, and inferential paradigms (risk factor analysis, causal evaluation, and prediction)
10/2024 – 12/2024	Lead Teaching Assistant , Epidemiologic Methods II Managed course operations, reviewed labs, and supported the teaching team for a class of 200+ graduate students covering causal inference, study design, and bias
01/2024 – 03/2024	Lead Teaching Assistant , Epidemiologic Methods III Managed course operations, reviewed labs, and supported the teaching team for a class of 200+ graduate students covering regression, survival analysis, and inferential paradigms (risk factor analysis, causal evaluation, and prediction)
01/2024 – 03/2024	Teaching Assistant , Spatial Statistics Led office hours and supervised labs in geostatistical, point-pattern, and area-level spatial statistical analysis
10/2023 – 12/2023	Teaching Assistant , Epidemiologic Methods II Provided in-person instruction in weekly labs and office hours covering causal inference, study design, and bias
10/2023 – 12/2023 10/2022 – 12/2022	Teaching Assistant , Spatial Analysis II: Spatial Data Technologies Led office hours and graded assignments covering a variety of tools for collecting, obtaining, and creating spatial data
03/2023 – 05/2023	Teaching Assistant , Foundations of Social Epidemiology Facilitated online discussion and gave individualized assignment feedback to strengthen students' knowledge of social epidemiologic concepts
08/2022 – 10/2022	Teaching Assistant , Spatial Analysis I: ArcGIS Led office hours and graded assignments covering the technical and substantive applications of geographic information systems for public health
2021 – Present	Reviewer , various journals Peer-review papers for <i>Preventing Chronic Disease</i>, <i>Health & Place</i>, and <i>Journal of Public Health Management & Practice</i>
2019 – 2022	Reviewer , ChristianaCare Way Awards

- Reviewed and critiqued draft project submissions for the ChristianaCare Way Awards, a competition for health system quality improvement projects

2018 – Present	Guest Lecturer , Johns Hopkins University, Thomas Jefferson University, and University of Delaware • Deliver guest lectures to graduate students on the use of geographic information systems and spatial statistics for public health research
06/2020 – 08/2020	Research Mentor , Delaware INBRE Summer Scholars Program • Supervised an undergraduate intern on a qualitative study of hospital registrars' attitudes and practices for collecting patient gender identity data
06/2018 – 01/2019	Research Mentor , Thomas Jefferson University • Mentored graduate public health students as they prepared manuscripts on the use of geographic information systems to study public health issues in Delaware; co-authored 3 manuscripts in the <i>Delaware Journal of Public Health</i>

PUBLICATIONS

Brooks MM, Gross AL, Lesko CR, Rubin LH, Akgun K, Hyle EP, Lang R, Amjad H, Napravnik S, Crane H, Lam JO, Piggott DA, Marconi V, Horberg M, Moore R, Althoff KN. Prescribing patterns of dementia medication in US persons with HIV and dementia diagnoses, 2016-2022. Under review.

Brooks MM, Gross AL, Lesko CR, Rubin LH, Lam JO, Napravnik S, Akgun K, Lang R, Hyle EP, Horberg M, Piggott DA, Althoff KN. Clinically recognized dementia incidence among US persons with HIV, 2016-2022. Under review.

Rashidi E, **Brooks M**, Hassoon A, Mehta S, Althoff K, Alexander GC. Is artificial intelligence a friend or foe to epidemiology? *Ann Epidemiol*. 2026;115:2-7. doi: <https://doi.org/10.1016/j.annepidem.2026.01.003>.

Brooks MM, Salvatore AL, Khanal P, Curriero FC. Mapping cumulative risk in Delaware: Approach and implications for health equity. *J Public Health Manag Pract*. 2024;30(3):E112-E123. [10.1097/PHH.0000000000001859](https://doi.org/10.1097/PHH.0000000000001859).

Poteat T, Hall P, **Brooks M**, Horn J, Yang C, Pereira N, Adams MA. Caregiving during the COVID-19 pandemic: A cross-sectional study with older cisgender sexual minority women in the United States. *LGBT Health*. 2024;11(3):219-228. <https://doi.org/10.1089/lgbt.2023.0226>.

Siegel SD, **Brooks MM**, Berman JD, Lynch SM, Sims-Mourtada J, Schug ZT, Curriero FC. Neighborhood factors and triple negative breast cancer: The role of cumulative exposure to area-level risk factors. *Cancer Med*. 2023;12:11760–11772. <https://doi.org/10.1002/cam4.5808>.

Bourke J, **Brooks MM**, Siegel SD. Policy recommendations for reducing tobacco exposure for youth and adults in Wilmington, Delaware. *Delaware J Public Health*. 2022;8(3):20-23. <https://doi.org/10.32481/djph.2022.08.006>.

Siegel SD, **Brooks MM**, Lynch SM, Sims-Mourtada J, Schug ZT, Curriero FC. Racial disparities in triple negative breast cancer: Toward a causal architecture approach. *Breast Cancer Res*. 2022;24(1):37. <https://doi.org/10.1186/s13058-022-01533-z>.

Mehta S, Waad A, **Brooks M**, Siegel SD. “I think there’s only two fields for that”: Hospital registrar attitudes and practices for collecting patient gender identity data. *Transgender Health*. 2023;8(4):400-403. <https://doi.org/10.1089/trgh.2021.0083>.

Brooks MM, Siegel SD, Corrigan AE, Curriero FC. Aggregated spatial intensity as a method for estimating point-level exposures within area-level units: The case of tobacco retailer exposure in census tracts. *Spat Spatiotemporal Epidemiol*. 2022;41:100482. <https://doi.org/10.1016/j.sste.2022.100482>.

Siegel SD, **Brooks MM**, Sims-Mourtada J, Schug ZT, Leonard DJ, Petrelli N, Curriero FC. A population health assessment in a community cancer center catchment area: Triple negative breast cancer, alcohol use, and obesity in New Castle County, Delaware. *Cancer Epidemiol Biomarkers Prev*. 2022;31(1):108-116. <https://doi.org/10.1158/1055-9965.EPI-21-1031>.

Harrison C, **Brooks M**, Goldstein J, Papas M. Food insecurity in Delaware: A triangulation of spatial data sources. *Prev Chron Dis*. 2021;18:E81. <https://doi.org/10.5888/pcd18.200555>.

Siegel SD, **Brooks M**, Bourke J, Curriero FC. Reducing exposure to tobacco retailers with residential zoning policy: insights from a geospatial analysis of Wilmington, Delaware. *Cities & Health*. 2022;6(4):752-764. <https://doi.org/10.1080/23748834.2021.1935141>.

Siegel SD, **Brooks MM**, Curriero FC. The co-occurrence of smoking and alcohol use disorder in a hospital-based population: Applying a multimorbidity framework using geographic information system methods. *Addict Behav*. 2021;118:106883. <https://doi.org/10.1016/j.addbeh.2021.106883>.

Brooks MM, Siegel SD, Curriero FC. Characterizing the spatial relationship between smoking status and tobacco retail exposure: Implications for policy development and evaluation. *Health & Place*. 2021; 68:102530. <https://doi.org/10.1016/j.healthplace.2021.102530>.

Siegel SD, **Brooks MM**, Curriero FC. Operationalizing the population health framework: Clinical characteristics, social context, and the built environment for a hospital-based population of current smokers. *Popul Health Manag*. 2021;24(4):454-462. <https://doi.org/10.1089/pop.2020.0170>.

Fonge YN, Jain VD, Harrison C, **Brooks M**, Sciscione AC. Examining the relationship between food environment and gestational diabetes. *Am J Obstet Gynecol MFM*. 2020;2(4):100204. <https://doi.org/10.1016/j.ajogmf.2020.100204>.

Brooks M, Brown C, Liu W, Siegel S. Mapping the ChristianaCare response to COVID-19: Clinical insights from the Value Institute's Geospatial Analytics Core. *Dela J Public Health*. 2020;6(2A):66-70. <https://doi.org/10.32481/djph.2020.07.018>.

Harrison C, **Brooks M**, Goldstein J. Social determinants of health, from assessment to action: A review of three studies from the Value Institute at ChristianaCare. *Dela J Public Health*. 2020;6(2A):72-77. <https://doi.org/10.32481/djph.2020.07.019>.

Siegel SD, **Brooks MM**, Gbadebo BM, Laughery JT. Using geospatial analyses of linked electronic health records and tobacco outlet data to address the social determinants of smoking. *Prev Chronic Dis*. 2019;16:E152. <https://doi.org/10.5888/pcd16.190186>.

Brooks M, Vest MT, Shapero M, Papas M. Malnourished adults' receipt of hospital discharge nutrition care instructions: a pilot study. *J Hum Nutr Diet*. 2019;32(5):659-666. <https://doi.org/10.1111/jhn.12662>.

McIntire RK, **Brooks M**, Siegel SD, Papas M. Mapping the burden of chronic disease in Delaware for public health decision making. *Dela J Public Health*. 2019;5(1):26-28. <https://doi.org/10.32481/djph.2019.02.007>.

Vecchione R, Dafilou C, Gerzen-Feschenko V, LaRatta N, **Brooks M**. Geographic patterns of asthma rates and air quality in Delaware. *Dela J Public Health*. 2019;5(1):30-37. <https://doi.org/10.32481/djph.2019.02.009>.

Fossi A, McCaffery D, Riseborough C, Vedhery N, Armstrong A, **Brooks M**. Food access in Delaware: examining the relationship of SNAP retailers, food deserts, obesity, and food insecurity. *Dela J Public Health*. 2019;5(1):58-65. <https://doi.org/10.32481/djph.2019.02.012>.

Sulaimon S, Smith R, Paz A, **Brooks M**. Using geographic information systems to display spatial patterns of diabetes in Delaware. *Dela J Public Health*. 2019;5(1):66-74. <https://doi.org/10.32481/djph.2019.02.013>.

McGhee J, Chambers PC, Cherico A, Bradley KM, **Brooks M**, Papas MA, Pirrung J, Sixta SL. Impact of a visible stopwatch on time that patients spend in the trauma bay during resuscitation. *Ann Emerg Med*. 2018;72(4):S114. <https://doi.org/10.1016/j.annemergmed.2018.08.294>.

Bouchaud MT, **Brooks M**, Swan BA. A retrospective analysis of nursing students' clinical experience in an all-male maximum security prison. *Nurse Educ*. 2018;43(4):210-214. <https://doi.org/10.1097/nne.0000000000000467>.

PRESENTATIONS

Brooks MM. A novel machine learning approach to identify neighborhood-level social phenotypes. Oral abstract presented during “Modern social epidemiology: advancing the field through innovative methodologies applied to community settings” at the Society for Epidemiologic Research Annual Meeting; June 23-26, 2026, Phoenix, AZ.

Brooks MM, Gross AL, Lesko CR, Rubin LH, Lam JO, Napravnik S, Piggott DA, Karris M, Akgun K, Horberg M, Althoff KN. Prescribing patterns of dementia-specific medication in US persons with HIV and clinically recognized dementia, 2016-2022. Poster presented at the Society for Epidemiologic Research Annual Meeting; June 23-26, 2026, Phoenix, AZ.

Brooks MM, Gross AL, Lesko CR, Rubin LH, Lam JO, Napravnik S, Eron J, Karris M, Horberg M, Kassaye S, Martin J, Piggott DA, Moore R, Althoff KN. Clinically recognized dementia incidence in US persons with HIV, 2016-2022. Poster presented at the International Workshop on HIV and Hepatitis Observational Databases; March 26-28, 2026, Thessaloniki, Greece.

Brooks M, Reid K, Zheng C, Zhao Yao, Humes E, Xu Y, Althoff K. Do machine learning and traditional regression models identify the same influential risk factors? A case study of virologic failure in people with HIV. Poster presented at Johns Hopkins Research Symposium on Engineering in Healthcare; December 8, 2025, Baltimore, MD.

Brooks MM. Research pearl: How to build a study flow chart. Presentation given at Welch Center Grand Rounds, Johns Hopkins Welch Center for Prevention, Epidemiology, and Clinical Research; September 10, 2025; Baltimore, MD.

Brooks M, Zhao Y, Zheng C, Humes E, Reid K, Moore R, Xu Y, Althoff A. Can random forests and deep neural networks add something new to our understanding of correlates of virologic failure in adults with HIV? A multidisciplinary team and a comparison of approaches. Poster presented at Johns Hopkins Bloomberg School of Public Health AI Research Day; November 13, 2024; Baltimore, MD.

Brooks MM. Research pearl: Using tmap to map area-level data. Presentation given at Welch Center Grand Rounds, Johns Hopkins Welch Center for Prevention, Epidemiology, and Clinical Research; September 18, 2024; Baltimore, MD.

Brooks MM. Research pearl: Using tidycensus to extract area-level data. Presentation given at Welch Center Grand Rounds, Johns Hopkins Welch Center for Prevention, Epidemiology, and Clinical Research; September 11, 2024; Baltimore, MD.

Brooks MM, Siegel SD, Curriero FC. Aggregated intensity: An extension of choropleth mapping. Poster presented at: Johns Hopkins GIS Day; November 16, 2022; Baltimore, MD.

Brooks M. Research highlights from the Johns Hopkins Spatial Science for Public Health Center: Spatial methods for characterizing tobacco retail exposure. Presentation given at: URISA GIS-Pro Conference; October 3-6, 2021; Baltimore, MD.

Brooks M. Spatial analysis of COVID-19 testing: An example from New Castle County, DE. Presentation given at: Delaware CTR-ACCEL Innovative Discoveries Series; September 23, 2021; presented virtually.

Brooks M, Patel V. Telemedicine visits during the COVID pandemic & beyond: Analysis of geographic distribution and relationship to determinants of health care inequity. Presentation given at: Delaware CTR-ACCEL Community Research Exchange; February 18, 2021; presented virtually.

Brooks M. Lightning talk: Smoking status and tobacco retail exposure in New Castle County, DE. Presentation given at: Johns Hopkins GIS Week. November 17-19, 2020; presented virtually.

Langston MC, **Brooks MM,** Subedi K, Shapero ML, Storm B, Woods R, Papas M, Vest MT. Characteristics of ICU patients requiring more than 10 days of enteral nutrition. Abstract accepted at: American Thoracic Society Conference; August 5-10, 2020; presented virtually.

Harrison C, **Brooks MM,** Papas M. Geographic distribution of food insecure patients at ChristianaCare primary care clinics in Delaware. Abstract accepted at: AcademyHealth 2020 Annual Research Meeting; June 13-16, 2020; Boston, MA. Not presented due to COVID-19.

Brooks MM, Axe MM, Ortiz J, Papas MA. From data flaws to data-driven practices: using continuous evaluation and knowledge transfer to improve the collection of patient race, ethnicity, and language. Poster presented at: AcademyHealth 12th Annual Conference on the Science of Dissemination and Implementation in Health; December 4-6, 2019; Arlington, VA.

Brooks MM, Siegel SD. The development of a geospatial analytics core for population health research. Poster presented at: Delaware IDeAs Symposium; November 7, 2019; Newark, DE.

Siegel SD, **Brooks MM**, Gbadebo BM, Laughery JT. Current and former smokers reside in different neighborhoods: a population-level geospatial analysis. Poster presented at: Society of Behavioral Medicine 40th Annual Meeting; March 6-9, 2019; Washington, DC.

Bouchaud MT, **Brooks MM**. Prisons as learning environments for nursing and public health practice. Poster presented at: Drexel University 2nd Urban Health Symposium; September 7-8, 2017; Philadelphia, PA.